

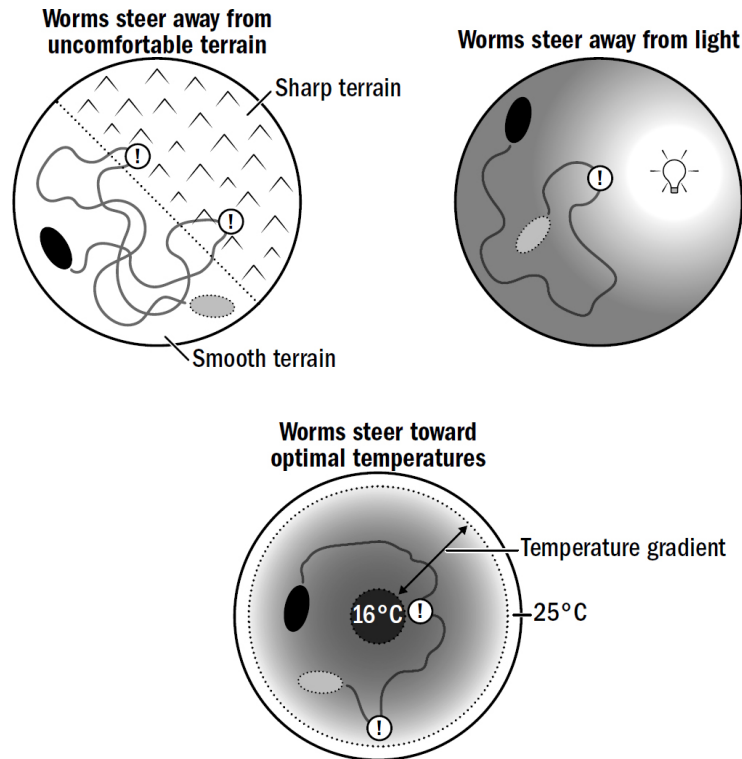


Figure 2.7: Examples of steering decisions made by simple bilaterians like nematodes and flatworms.

Original art by Rebecca Gelernter

The First Robot

In the 1980s and 1990s a schism emerged in the artificial intelligence community. On one side were those in the symbolic AI camp, who were focused on decomposing human intelligence into its constituent parts in an attempt to imbue AI systems with our most cherished skills: reasoning, language, problem solving, and logic. In opposition were those in the behavioral AI camp, led by the roboticist Rodney Brooks at MIT, who believed the symbolic approach was doomed to fail because “we will never understand how to decompose human level intelligence until we’ve had a lot of practice with simpler level intelligences.”

Brooks’s argument was partly based on evolution: it took billions of years before life could simply sense and respond to its environment; it took another five hundred million years of tinkering for brains to get good at motor skills and navigation; and only after all of this hard work did

language and logic appear. To Brooks, compared to how long it took for sensing and moving to evolve, logic and language appeared in a blink of an eye. Thus he concluded that “language . . . and reason, are all pretty simple once the essence of being and reacting are available. That essence is the ability to move around in a dynamic environment, sensing the surroundings to a degree sufficient to achieve the necessary maintenance of life and reproduction. This part of intelligence is where evolution has concentrated its time—it is much harder.”

To Brooks, while humans “provided us with an existence proof of [human-level intelligence], we must be careful about what the lessons are to be gained from it.” To illustrate this, he offered a metaphor:

Suppose it is the 1890s. Artificial flight is the glamor subject in science, engineering, and venture capital circles. A bunch of [artificial flight] researchers are miraculously transported by a time machine to the 1990s for a few hours. They spend the whole time in the passenger cabin of a commercial passenger Boeing 747 on a medium duration flight. Returned to the 1890s they feel invigorated, knowing that [artificial flight] is possible on a grand scale. They immediately set to work duplicating what they have seen. They make great progress in designing pitched seats, double pane windows, and know that if only they can figure out those weird “plastics” they will have the grail within their grasp.

By trying to *skip* simple planes and directly build a 747, they risked completely misunderstanding the principles of how planes work (pitched seats, paned windows, and plastics are the wrong things to focus on). Brooks believed the exercise of trying to reverse-engineer the human brain suffered from this same problem. A better approach was to “incrementally build up the capabilities of intelligence systems, having complete systems at each step.” In other words, to start as evolution did, with simple brains, and add complexity from there.

Many do not agree with Brooks’s approach, but whether you agree with him or not, it was Rodney Brooks who, by any reasonable account, was the first to build a commercially successful domestic robot; it was Brooks who made the first small step toward Rosey. And this first step in the evolution of commercial robots has parallels to the first step in the evolution of brains. Brooks, too, started with steering.

In 1990, Brooks cofounded a robotics company named iRobot, and in 2002, he introduced the Roomba, a vacuum-cleaner robot. The Roomba was a robot that autonomously navigated around your house vacuuming the

floor. It was an immediate hit; new models are still being produced today, and iRobot has sold over forty million units.

The first Roomba and the first bilaterians share a surprising number of properties. They both had extremely simple sensors—the first Roomba could detect only a handful of things, such as when it hit a wall and when it was close to its charging base. They both had simple brains—neither used the paltry sensory input they received to build a map of their environment or to recognize objects. They both were bilaterally symmetric—the Roomba's wheels allowed it to go forward and backward only. To change directions, it had to turn in place and then resume its forward movement.



Figure 2.8: The Roomba. A vacuum-cleaning robot that navigated in a way similar to the first bilaterians.

Photograph by Larry D. Moore in 2006. Picture published on Wikipedia at <https://en.wikipedia.org/wiki/Roomba>.

The Roomba could clean all the nooks and crannies of your floor by simply moving around randomly, steering away from obstacles when it bumped into them, and steering toward its charging station when it was low on battery. Whenever the Roomba hit a wall, it would perform a random turn and try to move forward again. When it was low on battery, the Roomba searched for a signal from its charging station, and when it detected the signal, it simply turned in the direction where the signal was strongest, eventually making it back to its charging station.

The navigational strategies of the Roomba and first bilaterians were not identical. But it may not be a coincidence that the first successful domestic robot contained an intelligence not so unlike the intelligence of the first brains. Both used tricks that enabled them to navigate a complex world without actually understanding or modeling that world.

While others remained stuck in the lab working on million-dollar robots with eyes and touch and brains that attempted to compute complicated things like maps and movements, Brooks built the simplest possible robot, one that contained hardly any sensors and that computed barely anything at all. But the market, like evolution, rewards three things above all: things that are *cheap*, things that *work*, and things that are simple enough to be discovered in the first place.

While steering might not inspire the same awe as other intellectual feats, it was surely energetically cheap, it worked, and it was simple enough for evolutionary tinkering to stumble upon it. And so it was here where brains began.

Valence and the Inside of a Nematode's Brain

Around the head of a nematode are sensory neurons, some of which respond to light, others to touch, and others to specific chemicals. For steering to work, early bilaterians needed to take each smell, touch, or other stimulus they detected and make a choice: Do I approach this thing, avoid this thing, or ignore this thing?

The breakthrough of steering required bilaterians to categorize the world into things to approach (“good things”) and things to avoid (“bad things”). Even a Roomba does this—obstacles are bad; charging station when low on battery is good. Earlier radially symmetric animals did not navigate, so they never had to categorize things in the world like this.

When animals categorize stimuli into good and bad, psychologists and neuroscientists say they are imbuing stimuli with *valence*. Valence is the goodness or badness of a stimulus. Valence isn't about a moral judgment; it's something far more primitive: whether an animal will respond to a stimulus by approaching it or avoiding it. The valence of a stimulus is, of course, not objective; a chemical, image, or temperature, on its own, has no goodness or badness. Instead, the valence of a stimulus is *subjective*, defined only by the brain's evaluation of its goodness or badness.